

# YAMAVARAPU BHANODAI

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## Summary

Final-year B.Tech student specializing in Artificial Intelligence & Machine Learning, with hands-on experience translating AI research into production solutions. Worked on applied research in vision-language models and retrieval-augmented generation. Built a top-10 Smart Visual Assistant for the visually impaired using deep learning and vision-language models. Proficient in TensorFlow and spaCy. Demonstrated leadership as Head of Tech-Web at NGC Club, driving innovation and cross-functional collaboration.

## Education

**Vellore Institute of Technology, Amaravati**, B.Tech in CSE-Minors in AI | GPA-8.32 *Est.2026*

- Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing, Operating Systems, Software Engineering, Data Structures and Algorithms

**Kendriya Vidyalaya**, High School Education *Completed 2020*

## Research & Work Experience

**AI Intern at Zigguratss**, Delhi, DL *Jun 2024 — Aug 2024*

- Delivered a Google Lens-like system for identifying and recommending visually similar paintings using machine learning and image similarity techniques, achieving 90% retrieval accuracy across a dataset of several thousand artworks through optimized feature indexing.
- **Technologies Used:** CLIP (Contrastive Language-Image Pretraining), Image Processing, Data Analysis, SQL, Python, Flask, FAISS (Facebook AI Similarity Search)

**Research Project on Agriculture Robot**, VIT Amaravati *Jan 2023 — May 2025*

- Developed an object detection pipeline using YOLOv5 to identify and localize weeds, achieving 85% mAP on a dataset of 5,000+ annotated crop images, enhancing precision in autonomous field monitoring.

## Projects

**Sentiment-Driven Predictive Analytics in the Bitcoin Market** *Sep 2024*

- Implemented a time-series model incorporating social media sentiment, which identified 4 key indicators influencing Bitcoin price fluctuations, and reduced prediction error by 12% compared to baseline model
- Collected and analyzed financial data from social media and news sources to assess market sentiment.
- Implemented machine learning models, including LSTMs, to predict cryptocurrency price fluctuations.

**Project Doom** *Feb 2025*

- Created an AI-powered application enabling users to upload PDFs, extract text, and interact through a conversational interface.
- Utilized LangChain for document processing, Google Generative AI for embeddings, and Streamlit for UI.
- Implemented text chunking, preprocessing, and API-based query handling to optimize document retrieval.

**AI-Powered Stock Trading Agent** *Jun 2025*

- Built an agentic AI system using CrewAI to autonomously analyze financial news, fetch stock data, and make trading decisions.
- Designed multiple collaborating agents with specialized roles (e.g., sentiment analysis, price monitoring, trade execution).

## Technologies

**Technical Skills:** Python, Java, R, SQL, Machine Learning, Deep Learning, Agentic AI Frameworks, Sequential Models, NLP, Predictive Modelling, Computer Vision, Research Work, Excel, Data Visualization, Tableau, Data Analytics, TensorFlow.

**Certifications & Training:** Machine Learning A-Z, 100 Days Python BootCamp, Deep Learning with TensorFlow, Google's AI Cloud Courses, AWS Academy Cloud Foundations

**Awards:** Top 10 out of 300+ teams in Engineering Clinics Expo. Selected for Global Jury Validation among 200 entries, Best Project Award.